5 · Read road documents and quantities

SKILLEDTRADESBC · LINE A

A5 · Describe basic road and highway profiles

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A5.1 · Review categories of projects	“Sort use, ownership and authority”
A5.2 · Describe types of owners	“Sort use, ownership and authority”
A5.3 · Describe common types of asphalt work	“Describe the road form”
A5.4 · Review specifications for different types of roads	“Sort use, ownership and authority”, “Recalculate after the revision”
A5.5 · Describe cross section of road	“Choose the right view” through “Describe the road form”

Learning-task wording: SkilledTradesBC (2016, p. 24; PDF viewer p. 26). The activities develop classroom understanding; the answer guide records the evidence to check.

Identify the project before the specification

A project may be described by use—commercial, industrial or residential—or by jurisdiction and delivery, such as a provincial highway project. These categories answer different questions and can overlap. A private industrial site is different from a provincially administered highway, even if both use asphalt. Owners can include provincial, federal, municipal or private organizations. A funder, designer, contractor and owner may be different parties.

Fictional practice project R1 · Cedar Learning Access, drawing P-01 revision B. The fictional Cedar Learning Society owns a short access route on its own practice property. Valley Surface Ltd. is the fictional contractor. The classroom project brief classifies the work as a private facility access project. A fictional community fund contributes money but does not become the owner. There is no real construction instruction or permission attached to this case.

The practice document register contains P-01 revision B, a plan/profile/section sheet; Q-01 revision B, a quantity record; and SP-01 revision B, a teaching specification card. An earlier P-01 revision A is retained for comparison and marked superseded. In a real project, the contract and authorized document system determine applicable requirements; a familiar document title does not establish that it governs every road.

Document families include B.C.'s provincial highway specifications, **Master Municipal Construction Documents (MMCD)**, a local government's project requirements, and building-specific architectural or civil drawings and specifications. MMCD's publisher produces standardized municipal construction documents and updates. A municipal or private project may adopt particular documents through its contract; naming MMCD

does not establish that every edition or clause applies. Provincial specifications likewise need the applicable contract version and special provisions. Record the issuer, edition, project adoption and any authorized change before relying on a requirement.

Sort use, ownership and authority

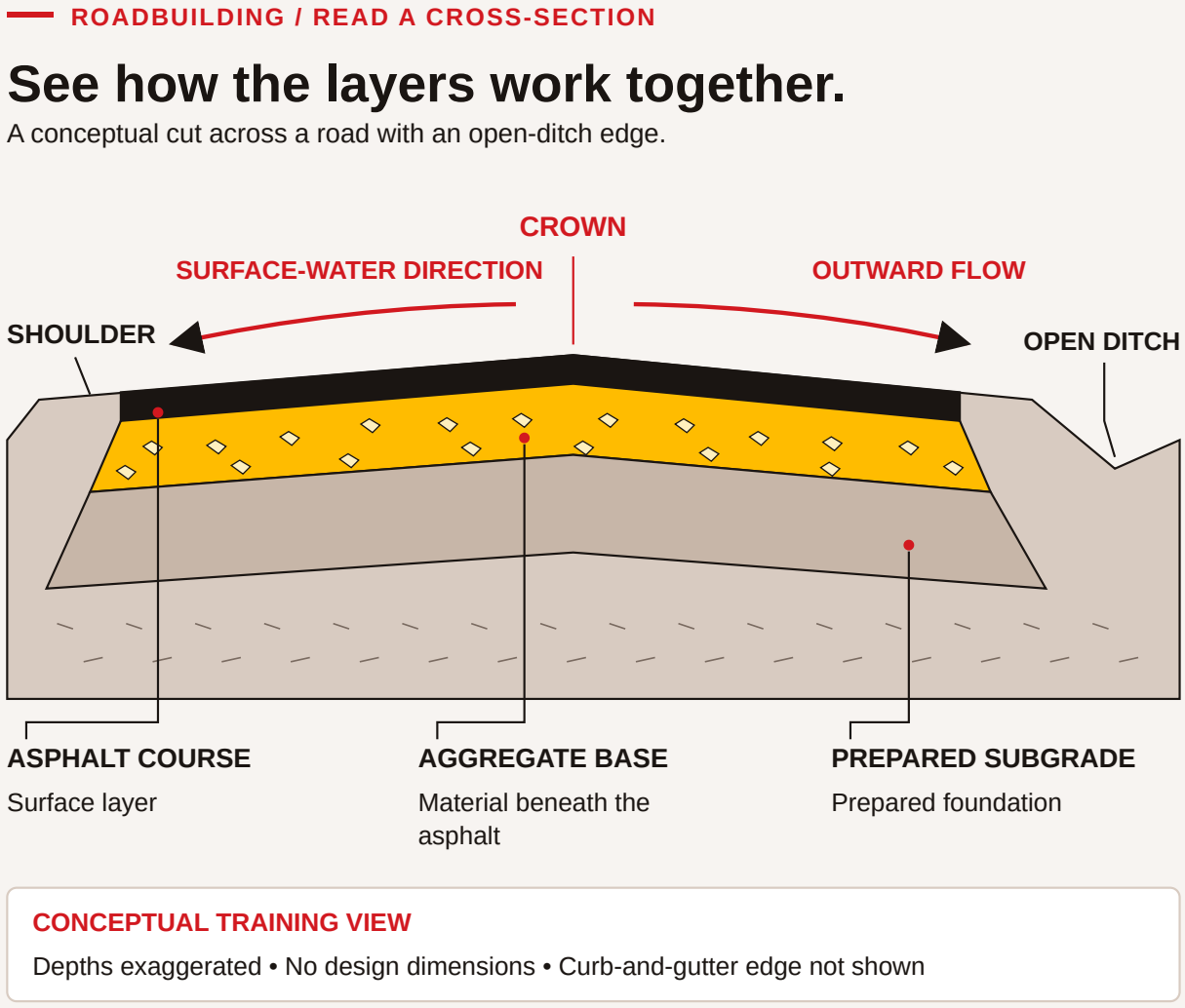
Identify the project use, owner, contractor and funder in R1. Explain why “private” and “asphalt” do not tell you which public standard or special provision applies. Compare provincial, MMCD, local-government and building-specific document families: what project-adoption evidence would you check before using any of them? Name the record you would need before treating a requirement as authorized. **Skills:** Reading, Problem Solving, Writing.

Read three views of the same place

A **plan** looks down on a project and shows its horizontal arrangement. A **profile** shows elevations along a chosen line, commonly using chainage or stationing to locate points. A **cross-section** cuts across the route and shows its transverse form and layers. A number must be associated with the correct view, direction and unit. Plan width, profile rise and layer thickness are different quantities.

Figure LA-2

See how the layers work together



Note. An original ACCESS IS4S instructional illustration. Layer depths are exaggerated. Curb-and-gutter and superelevation alternatives are discussed in the text, not drawn here. No design dimensions or acceptance standards are supplied. By ACCESS IS4S (2026b); see the figure references in the working reference section.

A **crown** is a raised transverse form that can direct water toward the sides. **Superelevation** describes a cross-slope arrangement used on curves; its design depends on the project. **Runoff** is water moving from the surface. An open ditch and shoulder arrangement differs from a contained curb-and-gutter edge. A sidewalk serves a different function from the vehicle surface. A grader is equipment used in earthwork or grading, not a layer in the finished section. Drainage and preparation must be coordinated before paving; a drawing needs more than a surface outline to establish a buildable design.

Distinguish the supporting layers

A **base** and **subbase** are supporting layers where the design includes them. The **prepared subgrade** is the prepared ground beneath the pavement structure. Figure LA-6 separates one unbound-base arrangement so you can name the layers. Other structures exist, including full-depth asphalt; a subbase is not automatically present on every road. This teaching illustration supplies no dimensions for P-01.

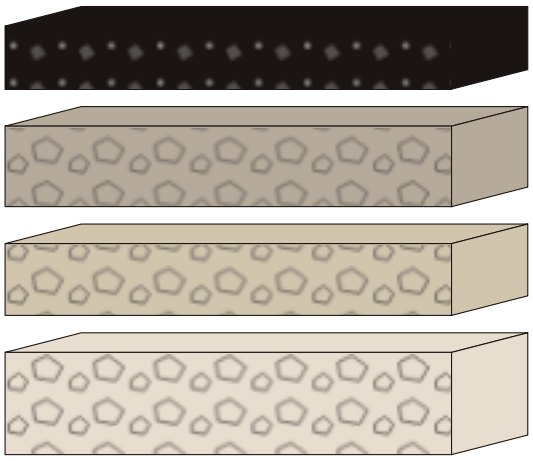
Figure LA-6

Name the layers

ACCESS IS4S · ASPHALT FOUNDATIONS

Name the layers

One example: asphalt pavement on an unbound base, with a subbase.



1 **Asphalt surface**
The upper asphalt layer.

2 **Granular base**
Supports the asphalt surface.

3 **Subbase, where specified**
An additional supporting layer.

4 **Prepared subgrade**
The prepared ground beneath the structure.

Conceptual section · other pavement structures exist · layer thicknesses are not a design.

Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 1.6, p. 14 / PDF 48), by Asphalt Institute, 2007. Selected the unbound-base example, separated its layers and added functional descriptions. Other structures and design dimensions are not implied. Copyright 2007 by Asphalt Institute; original contributor credits retained.

P-01 revision B · Complete fictional drawing data. The rectangle extends from chainage 0 m to 50 m. Its constant width is 6 m. The new uniform layer is 40 mm thick for this calculation exercise. No widening, taper, overlap, waste allowance or material loss is included. The exercise asks for geometric volume only. Use the written dimensions; a screen image is not a measuring scale.

Profile point	Chainage along centreline	Illustrative elevation
Start	0 m	100.00 m
Midpoint	25 m	100.25 m
End	50 m	100.50 m

The section names surface course, aggregate base and prepared subgrade. Their presence in a conceptual sketch does not specify material properties, layer thicknesses other than the given calculation layer, drainage design or acceptance criteria. All elevations and dimensions here are invented. The profile arithmetic is not a recommendation for road grade or crossfall.

Choose the right view

For route length, width, centreline rise and layer thickness, identify the relevant view and unit. Explain why the profile elevation 100.50 m is not the layer thickness. Sketch three boxes labelled plan, profile and cross-section and put the matching information in each. **Skills:** Reading, Numeracy, Creativity and Innovation.

Describe the road form

Using the visual and paragraph, explain crown, shoulder, open ditch, curb/gutter, sidewalk, runoff and superelevation. Put “grader” in the correct category. Use Figure LA-6 to distinguish granular base, subbase where specified and prepared subgrade; explain why one illustrated arrangement does not establish the layers on every project. Choose one drainage question and one preparation question that the conceptual drawing leaves unanswered. **Skills:** Reading, Communication, Problem Solving.

Point, name and explain

Use the drawing and the reading to connect each word to a function. A learner may point, annotate a copy, or describe the location verbally.

Trade word	Look for or explain	What the word does not tell you
Crown	Raised transverse form near the centre of a crowned road	The specified crossfall or an acceptance result
Shoulder / open ditch	Edge area / open drainage channel	Their required dimensions or capacity
Curb and gutter	A contained edge and associated drainage form	That it is drawn in this open-ditch example
Runoff	Surface water moving away from where it falls	The complete route, quantity or drainage design
Superelevation	A curve-related cross-slope arrangement	That it is the same shape as the crown in this figure
Grader	Equipment used in grading work	A layer in the cross-section
Prepared subgrade	The prepared foundation below the pavement structure	That its preparation has been inspected or accepted

Worked language check. “The arrows are runoff direction; the dark layer is the asphalt course.” The arrows describe movement; the layer describes material. In “Describe the road form”, explain two more features in the same way.

Calculate a quantity you can check

Worked example, P-01 revision B. Convert thickness first: `40 mm ÷ 1,000 = 0.040 m`. Plan area is `50 m × 6 m = 300 m²`. Geometric layer volume is `300 m² × 0.040 m = 12 m³`. Estimate: a 300 m² layer one-tenth of a metre thick would occupy 30 m³; a 0.04 m layer is less than half as thick, so 12 m³ is plausible. Units check as `m × m × m = m³`.

The profile rises `100.50 − 100.00 = 0.50 m` across 50 m of stated horizontal distance. For this drawing-reading exercise, rise/run is `0.50 ÷ 50 × 100 = 1%`. This centreline grade is not the transverse crown or an acceptance limit. A real quantity may need a different geometry and an approved measurement method;

our rectangular model has stated simplifications.

Q-01 revision B · Fictional mass model and delivery records. For this exercise only, the instructor supplies a conversion factor of **2.00 tonnes per cubic metre**. It is an invented teaching factor, not an asphalt density specification. The planned model mass is `12 m³ × 2.00 t/m³ = 24.00 t`. Three classroom records list 8.00 t, 8.00 t and 7.60 t. These are fictional record inputs, not a real delivery or placement measurement.

Check the quantity and record

Independently reproduce the area, thickness conversion, volume and model mass with units. Sum the three records and calculate their difference from the planned model mass. Explain why that numerical difference alone cannot establish placed thickness, compaction, payment entitlement or rejected work. **Skills:** Numeracy, Reading, Writing.

Use the applicable version and conditions

SP-01 revision B · Teaching specification card, fictional. Use the dimensions printed in the current classroom drawing. Record the revision with every calculation. Keep calculation quantities distinct from acceptance measurements. Refer conflicting instructions to the instructor acting as document coordinator before issuing a final classroom quantity note. This is the complete specification for the exercise; it does not adopt a real road specification.

Change notice R2, fictional. The coordinator issues P-01 revision C, extending the rectangle to 60 m while retaining 6 m width and 40 mm layer thickness. The endpoint illustrative elevation becomes 100.60 m; the same linear 1% profile is used for arithmetic practice. The mass model remains 2.00 t/m³. The three existing classroom records stay unchanged. Revision C replaces B for the new quantity calculation; retain B as the previous version.

Recalculate after the revision

Find the new area, volume and model mass, and the increase from revision B. Reconcile the unchanged record total with the new model requirement. Explain why changing a filename without recalculating would leave misleading information. Make a versioned paper or digital quantity note with inputs, method, check, result and unresolved limits. **Skills:** Numeracy, Adaptability, Writing; Digital when performed.

Review and hand over

Work in pairs: one checks geometry and units, the other checks document revision and claims. Swap responsibilities on the revised case. Give a handover that states the current revision, quantity, supplied factor, record total and the decision that remains outside this exercise. Each learner keeps a checked calculation and records one useful correction or confirmed detail. **Skills:** Collaboration, Communication, Problem Solving.

6 · Investigate a paving trend

SKILLEDTRADESBC · LINE A

A6 · Describe future paving/construction trends

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A6.1 · Describe green roads	“Improve three claims”, “Compare options against the actual need”
A6.2 · Describe new paving materials	“Make a product enquiry” through “Compare options against the actual need”
A6.3 · Describe new sub-bases	“Read the cost boundary”
A6.4 · Describe floating roads	“Read the cost boundary”, “Compare options against the actual need”
A6.5 · Describe added value enhancements	“Make a product enquiry”, “Build an evidence brief and test its clarity”

Learning-task wording: SkilledTradesBC (2016, p. 25; PDF viewer p. 27). The activities develop classroom understanding; the answer guide records the evidence to check.

Turn a promising idea into a testable question

A trend is useful to investigate when it responds to a defined need. A product described as *green*, *new* or *innovative* still needs evidence. Ask what problem it addresses, how it works, where it has been studied, what limits apply and who would approve its use. Something new to a local project may have been used elsewhere for decades. Publication date and location matter.

This unit compares material and design ideas without selecting a live road design. You will prepare a recommendation for further investigation using complete evidence cards and a fictional project brief. A careful recommendation can say that one option deserves a closer technical study while leaving design approval to the appropriate professional.

Keep the outline term and explain the mechanism

Outline wording	Meaning to investigate in this unit	First useful question
Green roads	A claim about an environmental purpose or outcome	Which outcome, comparison and time period?
Pave stones / paving stones	Individual paving units in a designed system	Does this particular system manage water through its joints?
Porous asphalt	A surface with interconnected voids, used within an appropriate pavement system	Where will water go, and how will the system be maintained?
Modified asphalt	Asphalt whose binder or mixture has been changed for a defined purpose	Which property changes, and what evidence supports the use?

Outline wording	Meaning to investigate in this unit	First useful question
New sub-bases / pumice	Foundation or fill ideas whose actual design role must be established	Is this material proposed as sub-base, embankment fill or something else?
Floating roads / Styrofoam	Historical outline wording discussed here through engineered lightweight fill and EPS geofoam	What engineered material and geotechnical design are intended?
Street print	A patterned surface treatment or visual enhancement	What are its purpose, performance and maintenance requirements?

A trade term helps you find the right subject. The project documents establish the actual material and use. Keep *sub-base* and *embankment fill* distinct when interpreting the historical cost example below.

Compare surface and reuse approaches

Trend card N1 · Paving stones. Modular paving uses individual units rather than one continuous asphalt layer. Some systems are designed to manage water through joints and underlying layers; the word *paver* alone does not establish permeability. Ask about the whole pavement system, loading, drainage, accessibility and maintenance. A patterned appearance is not environmental evidence.

Trend card N2 · Porous asphalt. FHWA's April 2015 technical brief describes porous asphalt pavements with stone reservoirs as systems intended to provide a surface and manage stormwater. Water moves through interconnected surface voids into the underlying reservoir. This differs from an open-graded friction course on a dense pavement, where water moves through the surface course and out toward the side. The brief identifies potential clogging and the need for specialized maintenance, as well as site/design limitations. A material name does not guarantee infiltration at an unsuitable site.

Trend card N3 · Recycling. Recovered asphalt may enter an approved plant or specialist in-place process. Recovered concrete may be processed as aggregate. Potential reductions in new-material demand need to be examined alongside processing, transport, suitability, maintenance and end-of-life. “Recycled” is a property of the material pathway, not a complete environmental score.

Improve three claims

Rewrite these statements so they are bounded and useful: “All paving stones absorb rain”; “Porous asphalt never needs drainage planning”; “Any recycled material is automatically the lowest-impact choice.” For each, identify a mechanism, a condition and one question for a qualified reviewer. **Skills:** Reading, Problem Solving, Writing.

Ask what a material changes

Trend card N4 · Epoxy, colour and modified asphalt. Epoxy systems, coloured surfaces and modified binders are families of products and applications. Colour can serve appearance or wayfinding, but it does not by itself establish structural performance, skid properties, durability or accessibility. A modified binder may be selected for a particular performance requirement. The full product description, intended use, test evidence, compatible construction system and approved specification are needed before adoption. Do not turn a product family into one universal recipe.

Trend card N5 · Lightweight fill and "floating roads." Lightweight fill reduces the load imposed by an embankment compared with a heavier fill arrangement. Expanded polystyrene, or EPS, geofoam is one engineered lightweight-fill material. Its use beneath a road does not mean the road floats on open water. The soil, loading, stability, protection, groundwater and installation conditions require geotechnical design. A brand name or generic word such as *foam* is not a material specification. Pumice is another material encountered in lightweight-fill discussions, with properties and availability that must be checked for the actual source and use.

FHWA's **April 2017 Ground Modification Methods, Volume I** describes lightweight fills and EPS geofoam. It explicitly states that the publication is not a standard, specification or regulation. Its chapter explains that material properties vary and detailed design requires further testing and evaluation. The examples show that lightweight-fill ideas are not all recent inventions.

Evidence table N6 · One historical U.S. cost example. FHWA's 2017 manual reproduces a Milpitas, California project comparison credited to Cheng 2016. The total costs below include material, transport and longevity costs. They **exclude installation and contractor overhead and profit**. They are historical U.S. project figures, not current B.C. quotes. The table's dollar figures are retained in their source context; no currency conversion is supplied.

Material	Reported example total
Traditional soil fill	$563,000
Pumice rock	$633,000
EPS-block geofoam	$1,145,000
Expanded shale/clay	$490,000
Wood chips	$545,000
Tire-derived aggregate	$334,000

Worked example. In this table, pumice's reported total exceeds traditional soil fill by `$633,000 − $563,000 = $70,000`. That result answers a question about this historical comparison. It does not establish that pumice always costs $70,000 more, that the cheapest listed option meets a new site's design needs, or that the table includes every cost.

Read the cost boundary

Calculate the difference between EPS-block geofoam and pumice in N6. Name three reasons the table cannot be used as a current B.C. quotation. Explain why a lightweight-fill cost table does not by itself establish an approved pavement sub-base material. Explain why price ranking alone cannot establish technical suitability. Keep units and the word *historical* in your conclusion. **Skills:** Numeracy, Reading, Problem Solving.

Separate an enhancement from a requirement

Trend card N7 · Patterned surfaces. Street-print or other patterned treatments can add a visual effect to a pavement. A visual enhancement may serve a project purpose, but performance claims still need evidence. Ask about the actual treatment, friction, wear, maintenance, access needs and compatibility with the underlying system. A decorative pattern is not evidence of structural strengthening or permission to use a cultural design. Any community-specific image or knowledge requires appropriate permission.

Make a product enquiry

Choose N4 or N7 and draft five questions for a fictional supplier. Include intended use, product identity, evidence, maintenance and approval/permission. Make one question specific enough that a vague marketing answer would be recognizable. Keep the enquiry local; no message is sent. **Skills:** Writing, Communication, Creativity and Innovation.

Apply the evidence to a defined case

Fictional project N8 · A learning-centre courtyard. A facility team is considering a 600 m² courtyard used by pedestrians and occasional authorized service vehicles. The team wants to improve surface-water management and readable wayfinding. No geotechnical investigation, infiltration test, approved loading design, maintenance budget or cultural-design permission has been supplied. The project budget is not yet set. A promoter offers porous asphalt, coloured asphalt or an EPS-supported pavement as three “green” options. These options address different questions and are not interchangeable solutions.

Compare options against the actual need

Make a table for the three offers. State the problem each might address, evidence provided in N1–N7, missing project information and the reviewer needed. Select one or two ideas for further investigation, with reasons. Do not approve construction or claim a total environmental winner. Each learner contributes an evidence-based question before the group makes its shortlist. **Skills:** Problem Solving, Collaboration, Reading.

Adapt to a new constraint

The fictional facility manager says there is no commitment to specialized pavement-cleaning equipment and asks for a maintenance plan before further consideration. Revise your shortlist or investigation plan. Explain why this new fact is relevant to porous asphalt and why it does not automatically approve another option. **Skills:** Adaptability, Communication, Problem Solving.

Build an evidence brief and test its clarity

Create a one-page brief containing the project question, one proposed investigation, a supported benefit, a limitation, missing information and a next decision. Add source titles/dates from the research record and mark predictions as predictions. Save and reopen the file if using Digital. Ask a partner to identify what you are recommending and what you are leaving undecided; revise one unclear sentence. **Skills:** Writing, Digital, Communication, Creativity and Innovation.

Keep the claims you improved, historical comparison, supplier questions, option table and revised brief. A useful innovation record shows the question, evidence, trial or investigation needed and the limits of the conclusion.

Working reference and evidence record

Words used in this book

Term	Meaning in these lessons
Aggregate	Mineral material such as sand, gravel or crushed stone
Asphalt binder / asphalt cement	Binding material in an asphalt mixture; distinct from Portland cement
Batch plant	Production arrangement that makes identifiable batches
Drum-mix plant	Production arrangement using a continuous material/mixing process; layouts vary
RAP	Reclaimed asphalt pavement material considered for an approved reuse pathway
Hot-in-place recycling	Specialist process treating existing asphalt pavement on the road
Screed	Paver component that contributes to forming the placed mat
Compaction	Densification process performed according to the applicable approved procedure
Milling	Removal or profiling of existing material using suitable equipment
Chainage	Position measured along a defined project line
Plan / profile / cross-section	Top view / elevation along a line / cut across a route
Crown	Raised transverse road form that can direct surface water toward sides
Superelevation	Cross-slope arrangement associated with a curve; design is project-specific
Runoff	Water moving from a surface
Specification	Documented requirements; applicability depends on its legal/contractual context
Special provision	Project/contract-specific provision that may modify or add requirements
Revision	Identified version of a document
Functional comparison	Comparison based on an equivalent required service, period and conditions
Life-cycle boundary	Stages included and excluded in an environmental assessment
Percentage point	Difference between two percentages expressed on the same basis
EPS in the 2006 specifications	End Product Specification, explicitly defined in clause 502.03.01; identify the edition and context before applying the term
EPS geofoam	Expanded-polystyrene material for an engineered geotechnical application; an abbreviation's meaning still depends on context

Read the word, find the evidence, explain the limit

To practise a term, write the trade word, point to its feature or source statement, and explain the function in one plain sentence. Then say what the picture or statement cannot establish. Keep technical words intact while making the explanation easier to follow. For example: “The **asphalt course** is the surface layer shown here; this picture does not specify its required thickness.”

Record evidence without overstating it

For each selected activity, record: activity ID; date; task and document version; what you did; support used; result; check; feedback; revision; and next practice. Mark the observed skill. A group product needs an individual contribution record. An accurate calculation needs units and assumptions. A digital claim needs actual tool use. A future application is a plan until it has been observed.

Activity	Evidence kept	Support and feedback	Next practice
–––––	–––––	–––––	–––––
–––––	–––––	–––––	–––––
–––––	–––––	–––––	–––––

Sources and continuing questions

The research register provides public titles, issuers, dates, URLs, retained short excerpts and retrieval hashes. It distinguishes dated factual evidence from invented classroom inputs and records questions that need current local or technical confirmation. Use the register to check a claim, then keep the question and the evidence together. A link or a confident statement alone is not the complete answer.

Supplied technical document

British Columbia Ministry of Transportation, Construction and Maintenance Branch. *2006 Design Build Standard Specifications for Highway Construction*, adopted 20 February 2006. Section 502.03.01, PDF 297 / printed 1 of 22, defines End Product Specification (EPS). The supplied historical document is distinct from the current public specifications page below; its presence does not establish current contract applicability.

Public reference list

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The curated evidence register gives the teaching use and limits; exact selected excerpts and hashes support checking. The road-industry table extract preserves the nine source rows.

Trade-outline and figure references (APA 7)

SkilledTradesBC. (2016). *Asphalt paving/laydown technician program outline* (August 2016). Program outline, Line A, printed pp. 20–25 / PDF pp. 22–27.

ACCESS IS4S. (2026a). *Follow the material, step by step* [SVG instructional illustration]. Local illustration.

ACCESS IS4S. (2026b). *See how the layers work together* [SVG instructional illustration]. Local illustration.

Figures LA-1 and LA-2 are original ACCESS IS4S teaching illustrations. Figure LA-3 retains the manufacturer's component callouts and is accompanied by a selected text legend. The trade terms are connected to the supplied outline; source-specific technical claims retain their evidence cards and reference entries.

LeeBoy. (2026). *LeeBoy model 8680 commercial paver: Safety and operations manual* (Manual No. 1033428-SOP). https://www.leeboy.com/wp-content/uploads/2024/03/1033428-SOP-MANUAL-8680-REV-A_WEB.pdf

Figure LA-3 uses Figure 3-1, printed p. 3-7 / PDF p. 43. The manufacturer names shown beside the crop are a selected subset of the complete legend. The image provenance manifest records the original PDF hash, page and crop boundary.

Asphalt Institute. (2007). *The asphalt handbook* (7th ed., MS-4). Complete local reader.

Figures LA-4–LA-6 are ACCESS redraws of verified MS-4 figures. LA-4 uses Figures 7.2 and 7.4 (pp. 310, 312 / PDF 344, 346; original contributors Terex Roadbuilding and National Highway Institute). LA-5 uses Figure 8.11 and component descriptions (pp. 377–378 / PDF 411–412; original figure contributor Volvo Construction Equipment). LA-6 uses one unbound-base arrangement from Figure 1.6 (p. 14 / PDF 48). Each caption identifies the simplification. The figure provenance manifest retains the source identity, asset hash and contributor-credit page. The chapter map connects the verified book contents to the occupational outline.